

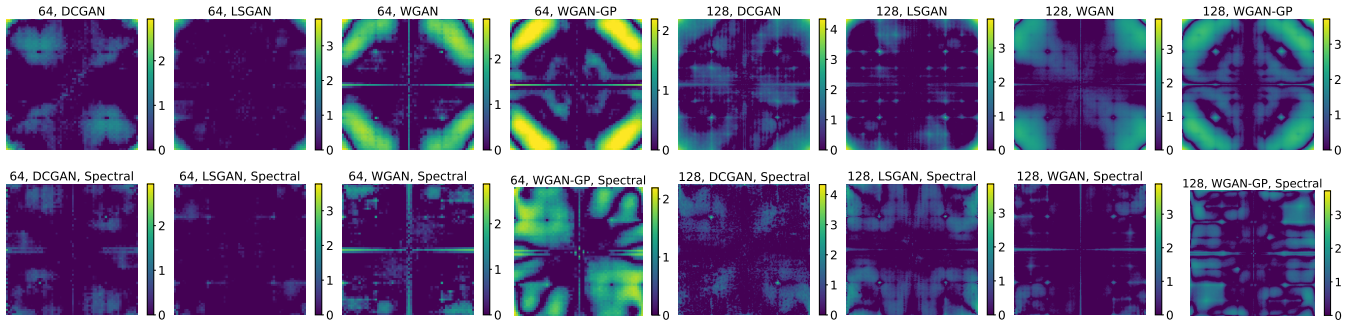


Figure 5: Mean absolute differences of the 2D power spectra of real and generated images. (Top) Differences for the original models. (Bottom) Differences for the models with spectral discriminator D_F . While the differences of the original models follow a characteristic pattern, our modified models show less specific differences and lower magnitudes.

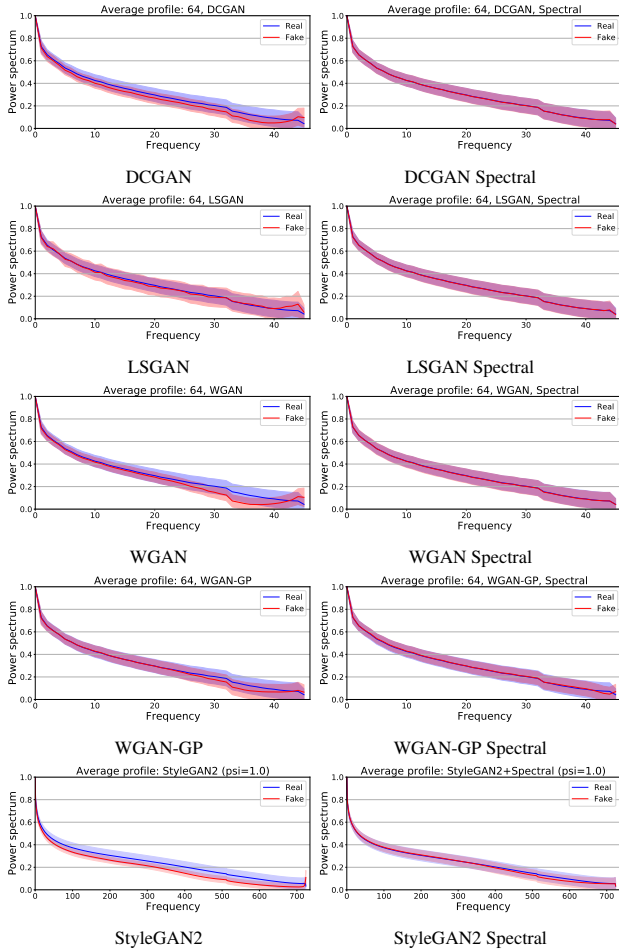


Figure 6: Resulting spectral profiles of experiments with our models trained on FFHQ64 and StyleGAN2. *Spectral* indicates that D_F was applied. With the spectral discriminator, the mean and standard deviation of the spectral profiles fit almost perfectly.

employs a simple discriminator on the 1D projections of fre-

quency spectra of real and generated images. Thus, the generator aims to match the real data distribution not only in the image domain but also in the frequency domain. Our approach is very lightweight and allows for a stable training, as we experimentally show for different loss functions and resolutions. It generates images that can not easily be distinguished from real ones by frequency artifacts and improves, in this respect, over the recent method by (?).

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